Experiment – 8:

Laplace Transform, Inverse Transform, ROC and Stability

(ii) Determination of Inverse Laplace Transform for Signal Reconstruction

Expected Outcomes

- Obtain x(t) from X(s).
- Display the inverse-transform expression.
- Evaluate the reconstructed waveform numerically.
- Analyze the resulting time-domain response.

Software / Laboratory Environment

MATLAB

Formulae / Analysis

X(s)=1/[(s+2)(s+3)].
Partial fractions: X(s)=1/(s+2)−1/(s+3).
x(t)=e^(−2t)−e^(−3t), t≥0.

Experimental Procedure

- Define X(s) symbolically.
- Apply ilaplace to obtain x(t).
- Evaluate x(t) over 0≤t≤5 s.
- Plot the reconstructed time-domain signal. Observe plot
- Compare the plotted response with the analytical expression.

Key Conditions & Threshold Values for Analysis

- Use Symbolic Math Toolbox.
- Use the specified time interval and step.
- The response should decay toward zero for the given poles.

MATLAB Coding

```
clc;
clear;
close all;

% Qualitative 8(ii): Determination of Inverse Laplace
% Transform for Signal Reconstruction

figure;

% Inverse Laplace Transform
% Requires Symbolic Math Toolbox

syms s t;

X = 1/((s+2)*(s+3));

x = ilaplace(X, s, t);

disp('Inverse Laplace Transform x(t) =');

pretty(x);

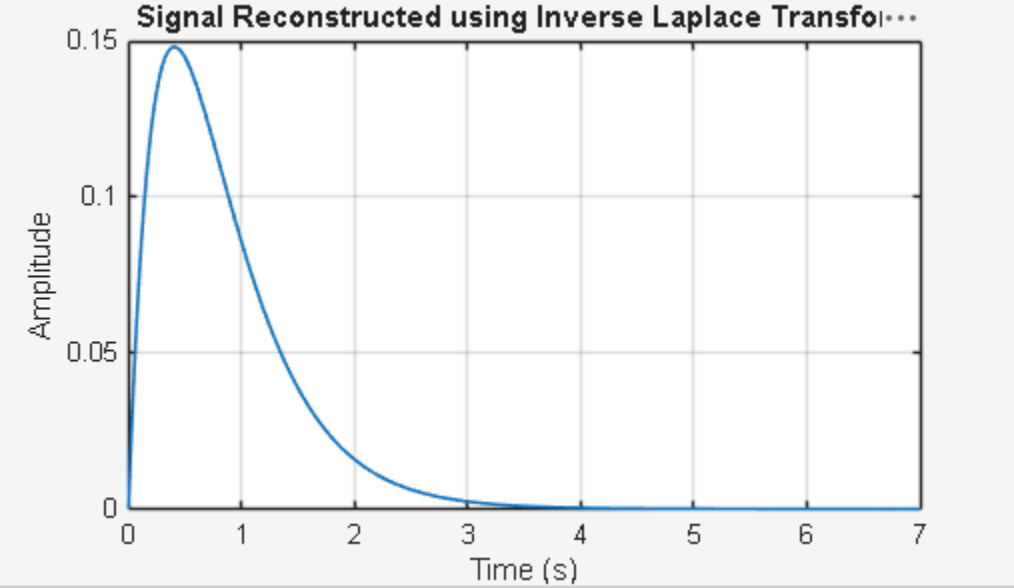
% Time range
tt = 0:0.001:7;

% Convert symbolic expression into numerical values
xx = double(subs(x, t, tt));
```

```
% Plot reconstructed signal
plot(tt, xx, 'LineWidth', 1.2);

grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Signal Reconstructed using Inverse Laplace Transform');
```

OUTPUT



Result:

The Laplace Transform and Inverse Laplace Transform were successfully studied and implemented using MATLAB. The Region of Convergence (ROC) was analyzed by identifying the poles of the system. The relationship between ROC and system stability was also understood. Thus, Laplace analysis helps in understanding system behaviour and stability.

4.

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage	Marks given	Marks obtained
(i) Determine the inverse Laplace transform of a given frequency-domain representation and reconstruct the corresponding time-domain signal.	**Inverse Laplace Transform Evaluation**	Correctly expands X(s) using partial fractions (distinct, repeated and complex poles) and applies the standard transform pairs and properties. Obtains the exact time-domain expression x(t), including the unit step u(t) and any shifts, with correct constants.	Applies partial fractions and transform pairs correctly, with minor errors in residues, constants or step-function terms.	Performs basic inverse transforms but has noticeable errors in partial fractions or transform pairs, and needs guidance.	Unable to determine the inverse Laplace transform.	**20**	**2**	
	Time-Domain Reconstruction & MATLAB Implementation	Uses suitable MATLAB commands (residue, ilaplace, syms, impulse or step) with a proper time vector. Reconstructs and plots x(t) with titles, axis labels and grid, and verifies the numerical result against the analytical expression.	Reconstructs and plots the signal correctly, with minor errors in code, time range, parameters or labelling.	Generates a basic reconstruction with noticeable errors in code or plots, and needs assistance.	Unable to reconstruct the time-domain signal or produce meaningful plots.	**20**	**2**	

(ii) Determine the Region of Convergence (ROC) of Laplace transforms and analyze its significance in characterizing continuous-time signals and systems.	**ROC Determination**	Correctly determines the ROC for right-sided, left-sided, two-sided and finite-duration signals from the pole locations. States the ROC as a proper inequality on Re(s) and shows it on the s-plane with poles and zeros marked.	Determines the ROC correctly for most signals, with minor errors in the boundary, signal type or s-plane plot.	Determines the ROC for basic cases only, with noticeable errors, and needs guidance.	Unable to determine the ROC of the given Laplace transforms.	**20**	**2**	
	ROC Significance for Signals and Systems	Clearly relates the ROC to the signal type, causality and BIBO stability (ROC contains the jω-axis), and shows that the same X(s) with different ROCs gives different time signals. Supports the reasoning with mathematical conditions and pole-zero evidence.	Relates the ROC to causality and stability correctly, with minor errors or limited justification.	Gives a basic explanation of the ROC's significance. Justification is incomplete and needs guidance.	Unable to explain the significance of the ROC.	**20**	**2**	
Interpret the Laplace-domain results and document the experiment.	**Engineering Analysis, Interpretation & Record**	Provides technically accurate interpretation of the results (pole locations, ROC and signal behaviour). Maintains a complete, well-organized	Maintains a proper record with generally accurate interpretation, with minor omissions.	Record and interpretation are basic and incomplete, with some technical inaccuracies.	Incomplete record, limited or incorrect interpretation of the results.	**20**	**2**	

		record with code, outputs, plots and conclusions.						
					Total	**100**	**10**	